

Fitting Covariance vs Response

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We fit connectivity in two ways: by fitting the raw responses, and fitting the covariance of the responses. How are the resulting connectivities related?

When we fit the responses, our loss is

$$\mathcal{L}_{\text{resp}}(\mathbf{Z}) = \frac{1}{2} \|\mathbf{Y} - \mathbf{Z}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2,$$

while when fitting covariances, we minimize

$$\mathcal{L}_{\text{cov}}(\mathbf{Z}) = \frac{1}{2} \|\mathbf{Y}^T \mathbf{J} \mathbf{Y} - \mathbf{X}^T \mathbf{Z}^T \mathbf{J} \mathbf{Z} \mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2.$$

To relate these, we use the decomposition into mean-subtracted and mean components.

$$\mathbf{X} = \mathbf{J}\mathbf{X} + (\mathbf{I} - \mathbf{J})\mathbf{X} \triangleq \tilde{\mathbf{X}} + \bar{\mathbf{X}}.$$

Notice that these two components are orthogonal,

$$\langle \tilde{\mathbf{X}}, \bar{\mathbf{X}} \rangle = \text{tr}(\tilde{\mathbf{X}}^T \bar{\mathbf{X}}) = \text{tr}(\mathbf{X}^T \mathbf{J}^T (\mathbf{I} - \mathbf{J}) \mathbf{X}) = \text{tr}(\mathbf{X}^T \mathbf{0} \mathbf{X}) = 0.$$

That means that we can decompose our response loss into two independent components,

$$\begin{aligned} \mathcal{L}_{\text{resp}}(\mathbf{Z}) = \mathcal{L}_{\text{resp}}(\mathbf{J}\mathbf{Z}, \bar{\mathbf{Z}}) &= \frac{1}{2} \|\mathbf{J}\mathbf{Y} - \mathbf{J}\mathbf{Z}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{J}\mathbf{Z} - \mathbf{J}\|_F^2 \\ &\quad + \frac{1}{2} \|\bar{\mathbf{Y}} - \bar{\mathbf{Z}}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\bar{\mathbf{Z}} - \bar{\mathbf{I}}\|_F^2. \end{aligned}$$

We can do the same thing for the covariance loss,

$$\begin{aligned} \mathcal{L}_{\text{cov}}(\mathbf{Z}) = \mathcal{L}_{\text{cov}}(\mathbf{J}\mathbf{Z}, \bar{\mathbf{Z}}) &= \frac{1}{2} \|(\mathbf{J}\mathbf{Y})^T \mathbf{J} \mathbf{Y} - \mathbf{X}^T (\mathbf{J}\mathbf{Z})^T (\mathbf{J}\mathbf{Z}) \mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{J}\mathbf{Z} - \mathbf{J}\|_F^2 \\ &\quad + \frac{\lambda}{2} \|\bar{\mathbf{Z}} - \bar{\mathbf{I}}\|_F^2. \end{aligned}$$

From this, we immediately see that the regularization in the covariance case will send $\bar{\mathbf{Z}}$ to $\bar{\mathbf{I}}$.

Therefore, the aspect of the connectivity to compare between the two losses is $\mathbf{J}\mathbf{Z}$. We can polar decompose this into the product of a pure rotation $\mathbf{R}$ and a positive-semi-definite stretch $\mathbf{S}$,

$$\mathbf{J}\mathbf{Z} = \mathbf{R}\mathbf{S}.$$

Since $(\mathbf{J}\mathbf{Z})^T (\mathbf{J}\mathbf{Z}) = \mathbf{S}^T \mathbf{R}^T \mathbf{R} \mathbf{S} = \mathbf{S}^T \mathbf{S}$, we can see that the covariance fit only determines the stretch component, while the response fit determines both.

Therefore, when comparing connectivity solutions, we should use $\mathbf{R}$ and $\mathbf{S}$ computed from $\mathbf{J}\mathbf{Z}$, not $\mathbf{Z}$, when determine the effects of each component.

Response Fits

Returning to the response fits, we find that Free fit, and its symmetrically constrained version, give the same performance. In fact, the symmetrically constrained one might be a little better. What do the fits look like?

The free fits is easy. The gradient is

$$\nabla \mathcal{L}_{\text{resp}} = -(\mathbf{Y} - \mathbf{Z}\mathbf{X})\mathbf{X}^T + \lambda(\mathbf{Z} - \mathbf{I}).$$

Setting this to zero gives

$$\mathbf{Z} = (\mathbf{Y}\mathbf{X}^T + \lambda\mathbf{I})(\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I})^{-1}.$$

If we define

$$\Sigma_{xx} \triangleq \mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}, \quad \Sigma_{xy} \triangleq \mathbf{X}\mathbf{Y}^T + \lambda\mathbf{I}, \quad \Sigma_{yx} \triangleq \mathbf{Y}\mathbf{X}^T + \lambda\mathbf{I},$$

we get

$$\mathbf{Z}_{\text{free}} = \Sigma_{yx}\Sigma_{xx}^{-1}.$$

This also gives

$$\mathbf{Z}_{\text{free}}^T = \Sigma_{xx}^{-1}\Sigma_{xy}^T = \Sigma_{xx}^{-1}\Sigma_{xy}.$$

To constrain the solution to be symmetric, we add a Lagrange multiplier to the loss, and get

$$\mathcal{L}_{\text{sym}} = \mathcal{L}_{\text{resp}} + \text{tr}(\Lambda^T(\mathbf{Z} - \mathbf{Z}^T)).$$

The gradient then becomes

$$\begin{aligned} \nabla \mathcal{L}_{\text{sym}} &= -(\mathbf{Y} - \mathbf{Z}\mathbf{X})\mathbf{X}^T + \lambda(\mathbf{Z} - \mathbf{I}) + \Lambda - \Lambda^T. \\ \nabla \mathcal{L}_{\text{sym}}^T &= -\mathbf{X}(\mathbf{Y}^T - \mathbf{X}^T\mathbf{Z}) + \lambda(\mathbf{Z} - \mathbf{I}) + \Lambda^T - \Lambda. \end{aligned}$$

Adding them when the gradient is zero eliminates the Lagrange multipliers and gives

$$\mathbf{X}\mathbf{Y}^T + \mathbf{Y}\mathbf{X}^T + 2\lambda\mathbf{I} = \mathbf{Z}\mathbf{X}\mathbf{X}^T + \mathbf{X}\mathbf{X}^T\mathbf{Z} + 2\lambda\mathbf{Z},$$

or, in terms of our definitions,

$$\Sigma_{xy} + \Sigma_{yx} = \mathbf{Z}\Sigma_{xx} + \Sigma_{xx}\mathbf{Z}.$$

Σ_{xx} is invertible, so we get equivalently

$$\Sigma_{xx}^{-1}\Sigma_{xy} + \Sigma_{xx}^{-1}\Sigma_{yx} = \Sigma_{xx}^{-1}\mathbf{Z}\Sigma_{xx} + \mathbf{Z},$$

or equivalently

$$\Sigma_{xx}^{-1}\Sigma_{xy}\Sigma_{xx}^{-1} + \Sigma_{xx}^{-1}\Sigma_{yx}\Sigma_{xx}^{-1} = \Sigma_{xx}^{-1}\mathbf{Z} + \mathbf{Z}\Sigma_{xx}^{-1}.$$

Then, using our expressions for the free solution, we get

$$\mathbf{Z}_{\text{free}}^T \Sigma_{xx}^{-1} + \Sigma_{xx}^{-1} \mathbf{Z}_{\text{free}} = \Sigma_{xx}^{-1} \mathbf{Z} + \mathbf{Z} \Sigma_{xx}^{-1},$$

or

$$\Sigma_{xx} \mathbf{Z}_{\text{free}}^T \Sigma_{xx}^{-1} + \mathbf{Z}_{\text{free}} = \mathbf{Z} + \Sigma_{xx} \mathbf{Z} \Sigma_{xx}^{-1}.$$

If we work in the eigenbasis of Σ_{xx} , using the tilde to indicate the coefficients, we can express those $\tilde{S}_{ij}$ of the symmetric solution in terms of those $\tilde{F}_{ij}$ of the free solution as

$$\tilde{S}_{ij} = \frac{\tilde{F}_{ji}D_i + \tilde{F}_{ij}D_j}{D_i + D_j},$$

where D_i are the eigenvalues of Σ_{xx} .

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What the data can see: the uniform direction

The decomposition above splits $\mathbf{Z}$ by what the covariance loss can see. There is a second, finer split, by what the DATA can see, and it explains why the symmetric fit behaves differently from the free one along one direction.

Per-odour normalization makes every column of $\mathbf{X}$ sum to zero over rois, $\mathbf{X}^T \mathbf{1} = \mathbf{0}$. Let $\mathbf{e} = \mathbf{1}/\sqrt{m}$ be the unit vector along the uniform direction and let $\mathbf{B}$ be any $m \times (m-1)$ matrix whose columns are an orthonormal basis of $\mathbf{1}^\perp$, so that $\mathbf{Q} = [\mathbf{e} \ \mathbf{B}]$ is orthogonal. Every $\mathbf{Z}$ then has the block form

$$\mathbf{Q}^T \mathbf{Z} \mathbf{Q} = \begin{bmatrix} a & \mathbf{r}^T \\ \mathbf{c} & \mathbf{M} \end{bmatrix}, \quad a = \mathbf{e}^T \mathbf{Z} \mathbf{e}, \quad \mathbf{c} = \mathbf{B}^T \mathbf{Z} \mathbf{e}, \quad \mathbf{r} = \mathbf{B}^T \mathbf{Z}^T \mathbf{e}, \quad \mathbf{M} = \mathbf{B}^T \mathbf{Z} \mathbf{B}.$$

Reading the blocks by what they map to what: a takes uniform input to uniform output, $\mathbf{c}$ takes uniform input to patterned output, $\mathbf{r}$ takes patterned input to uniform output – how strongly an input pattern drives the MEAN over rois – and $\mathbf{M}$ takes patterned input to patterned output. Nothing below depends on which basis $\mathbf{B}$ is: a , $\|\mathbf{c}\|$, $\|\mathbf{r}\|$ and the spectrum of $\mathbf{M}$ are invariant under $\mathbf{B} \rightarrow \mathbf{B}\mathbf{O}$ for orthogonal $\mathbf{O}$.

Because every column of $\mathbf{X}$ lies in $\mathbf{1}^\perp$, the product $\mathbf{Z}\mathbf{X}$ involves only $\mathbf{Z}\mathbf{B}$, which is to say only $\mathbf{r}$ and $\mathbf{M}$. So the response loss determines $\mathbf{r}$ and $\mathbf{M}$ and says nothing whatever about a or $\mathbf{c}$; the regularizer $\frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2$ sets those, giving $a = 1$ and $\mathbf{c} = \mathbf{0}$. The covariance loss sees even less: $\mathbf{J}$ annihilates the uniform component of the output, so $\mathbf{r}$ is invisible to it as well, and its regularizer sets $\mathbf{r} = \mathbf{0}$ too.

Imposing symmetry couples the two. $\mathbf{Z} = \mathbf{Z}^T$ forces $\mathbf{c} = \mathbf{r}$, so the data-determined $\mathbf{r}$ is mirrored into the $\mathbf{c}$ the data cannot constrain. This is not a failure of the optimizer: a still comes out at exactly 1, which is the one number that is genuinely free. It is the constraint acting on a block where the data is one-sided.

The consequence is visible in $\mathbf{Z}$'s spectrum, since

$$\mathbf{Z}\mathbf{1} = \sqrt{m}(a\mathbf{e} + \mathbf{B}\mathbf{c}) \implies \|\mathbf{Z}\mathbf{1} - \mathbf{1}\|^2 = m[(a-1)^2 + \|\mathbf{c}\|^2],$$

so a mirrored $\mathbf{c}$ moves $\mathbf{Z}$ away from fixing $\mathbf{1}$, and the eigenvectors that carry the uniform direction carry the effect. On the matched rois, the free response fit has $\|\mathbf{r}\| = 2.06$ with $\|\mathbf{c}\| = 0$, the covariance fit has both zero, and the symmetric refit has $\|\mathbf{c}\| = \|\mathbf{r}\| = 1.86$. The symmetric refit's two extreme eigenvalues, $+2.74$ and -2.09 , have overlaps of 0.67 and 0.50 with $\mathbf{e}$; restricted to $\mathbf{M}$ they are $+2.05$ and -1.23 .

The moral for the constrained refits – symmetric, PSD, rotation, orthogonal – is that a constraint imposed on $\mathbf{Z}$ reaches this block, where one side of it was never measured. A constraint imposed on $\mathbf{M}$ alone, leaving a , $\mathbf{c}$ and $\mathbf{r}$ to the regularizer, tests the same hypothesis about the part of the connectivity the data actually determines.